

Motivation

TTM Technologies relies on high-precision measurements to verify the quality of its printed circuit board (PCB) registration — the accuracy of PCB artwork layer positioning — at the start of every production batch. Currently, TTM assesses this measurement using an outdated machine that delivers unreliable and inconsistent results.

Objective

Design, manufacture, and program a modern Front-to-Back (F2B) measuring machine to replace the antiquated system. The new machine must accurately measure the translational offset of PCB layers to within ±0.1 mil, integrate with TTM’s database using Industry 4.0 standards, and feature a user-friendly Human-Machine Interface (HMI).

Selected Design Solution

Considered Factors	Methods Used	Solutions Considered
Camera Vision System Accuracy	Zebra: Individually Programmed machine vision system with telecentric lenses	Keyence: controller-based camera system with telecentric lenses Cognex: separate smart cameras with liquid lenses
Table Mounting System	Geometrically simple mounting system above/below center of table; adjustable 3D printed mounts	Complicated mounting system based over corner or multiple locations; Static purchasable mounting pieces
Electrical Box Layout	Three DIN rails with double layered terminal block connecting system; Camera communication through Ethernet switch; 24V cooling fan	Two/four DIN rails without terminal blocks; PC and controller located within electrical box; 120V industrial fan
HMI Operator Usability	Dual-visible live input feeds; Temporary data storage for current board; Pop-up instructions; Top-down input orientation; Monochrome minimalist visual design	Singular switchable live feed; Static stored data; visible instructions; Large input fields; Color-based visual inputs/outputs

Designs & Selected Components:

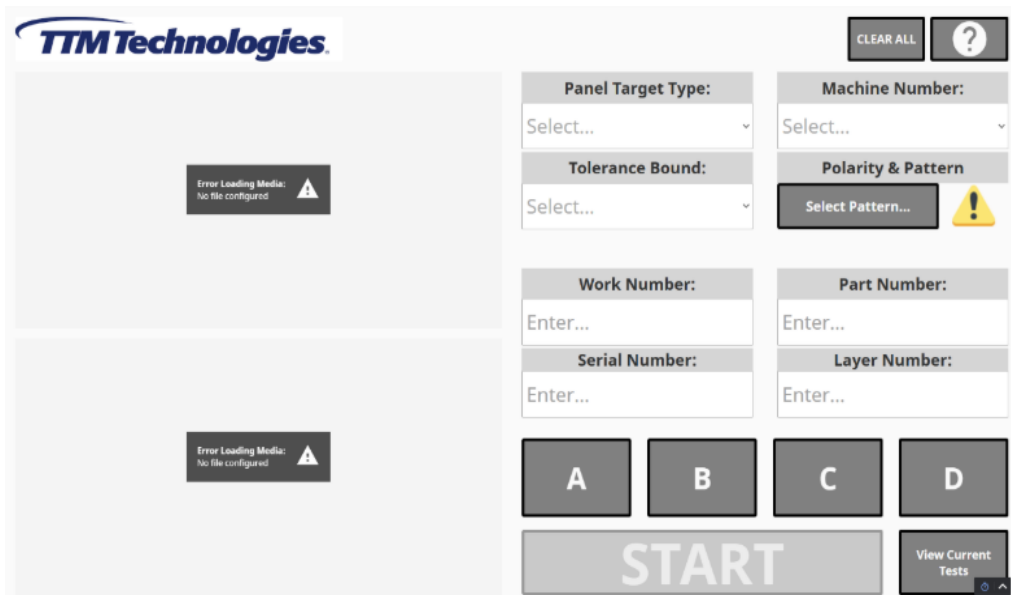


Figure 1: HMI Appearance

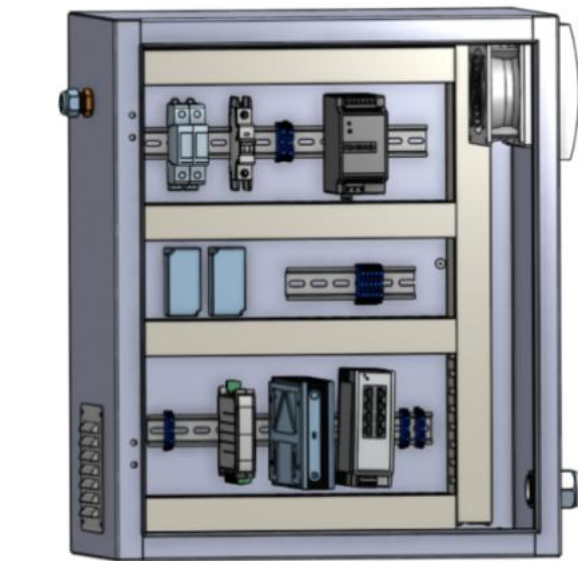


Figure 2: Electrical Box Interior

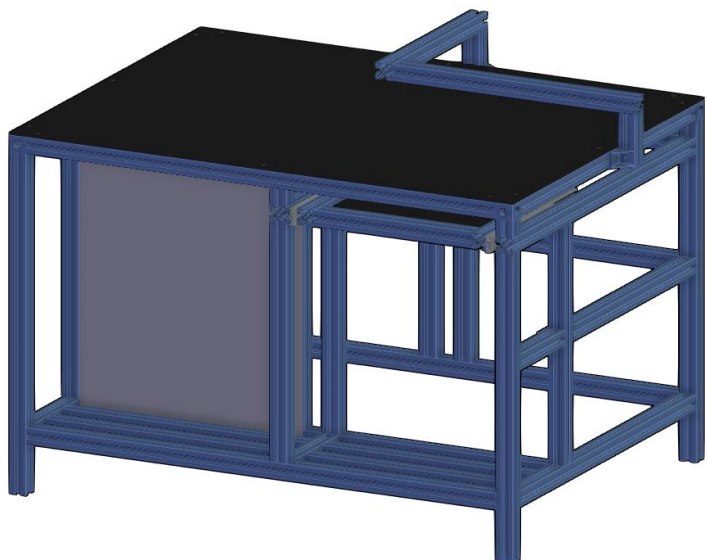


Figure 3: Table Design



Figure 4: Zebra Cameras

Specifications

F2B Inspection System Component Requirements	Target Specifications/Engineering Standards
Camera Vision System Accuracy	Consistent accuracy of ± 0.1 Mil (0.0001”) within 1/3” field of view (FOV)
Table Space Usage	Dimensions of 3’ x 4’ x 3’
Electrical Box Safety	Operates per National Electric Code (NEC) (NFPA70) rules for avoiding electrical fires
HMI Usability	Interface follows ISA101 standards for clear display of information

Proof of Concept Implementation

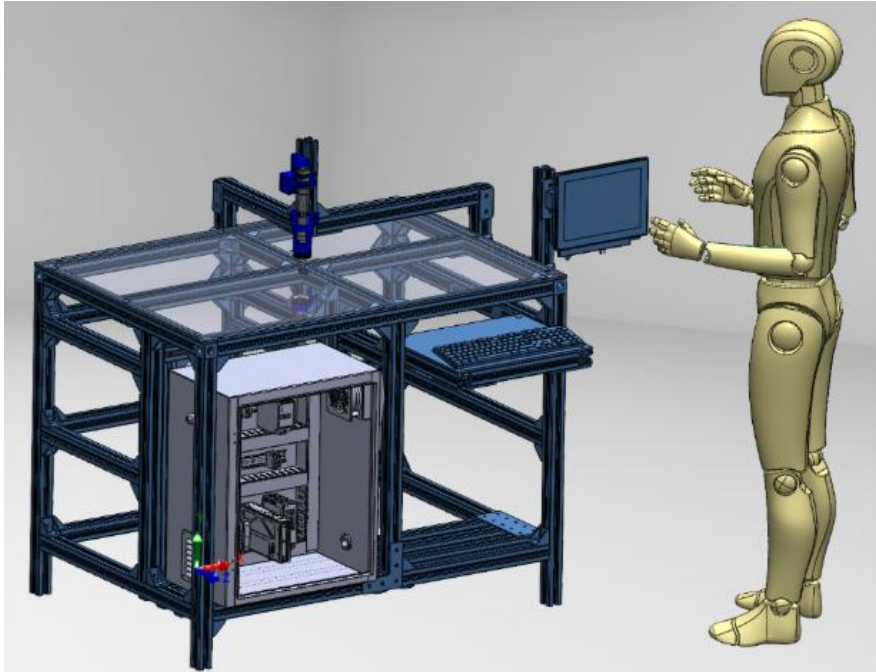


Figure 5: Full Proof of Concept Design

- Table designed in Vention
- Zebra cameras connected to mounting arms
- HMI displayed on mounted monitor
- Electrical Box (No Front Door)
- Keyboard Pullout Shelf
- Average Male (5’8”) for scale

Project #1407, F2B Inspection System

Michael Carlone – Design & Analysis Lead
Michael Grady – Systems Lead
Alex Henner – Fabrication Lead
Dinor Nallbani – Team Lead
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Supporting Engineering Analysis

Heat transfer and vibrational resistance analyses on the electrical box and table were done to ensure the design could handle worst-case scenario loads. Where necessary, designs were iterated to meet these requirements.

Heat Transfer Analysis (Electrical Box):

Determined **required** flowrate of the internal fan for the enclosure temperature **to not exceed 30°C**.

$$\dot{m} = \frac{Q_I}{c_p \Delta T} = 0.0100 \text{ kg/s}$$

$$\dot{V} = \frac{\dot{m}}{\rho} = 0.0082 \text{ m}^3/\text{s}$$

$$CFM_{required} \approx 17.89 \text{ CFM}$$

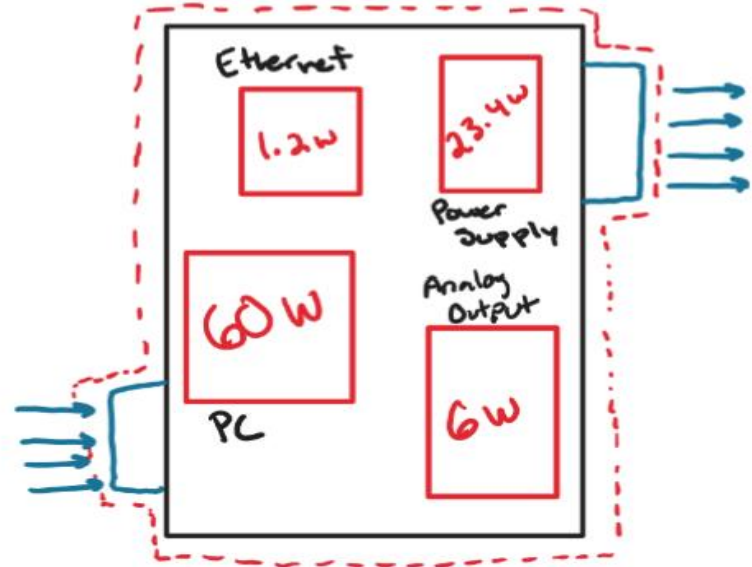


Figure 6: Electrical Box Control Volume

Vibration Analysis (Table + Arms):

Calculated Natural frequency **must not** exceed 100Hz to ensure no vibrations under assumed factory floor conditions.

Table (Iterations):

Configuration	Shelves	Panel	Total Mass <i>M</i> (kg)	<i>f</i> _{nat} (Hz)
Bare table (cantilever legs)	0	–	35.7	11.1
Full-foot rubber, one shelf	1	–	48.9	27.1
Full-foot rubber, two shelves	2	–	62.1	28.2
Two shelves + HDPE back panel	2	HDPE (top→lower shelf)	66.3	48.8
Two shelves + SS back panel	2	Stainless (top→first shelf)	70.9	181

Arms:

Arm 1 (1.5ft, 65N Cantilevered):

Arm 2 (2ft, 65N Cantilevered):

$$\delta_{max} = \frac{WL^3}{3EI} = 0.288 \text{ mm}$$

$$\delta_{max} = \frac{WL^3}{3EI} = 0.445 \text{ mm}$$

$$k_A = \frac{3EI}{L^3} \approx 302,377.55 \frac{N}{m}$$

$$k_B = \frac{3EI}{L^3} \approx 87,738.15 \frac{N}{m}$$

$$\omega_{n,a} = \sqrt{\frac{k_A}{m}} \approx 329.21 \text{ Hz}$$

$$\omega_{n,b} = \sqrt{\frac{k_B}{m}} \approx 177.33 \text{ Hz}$$

Performance Evaluation

Example Measurement Test:

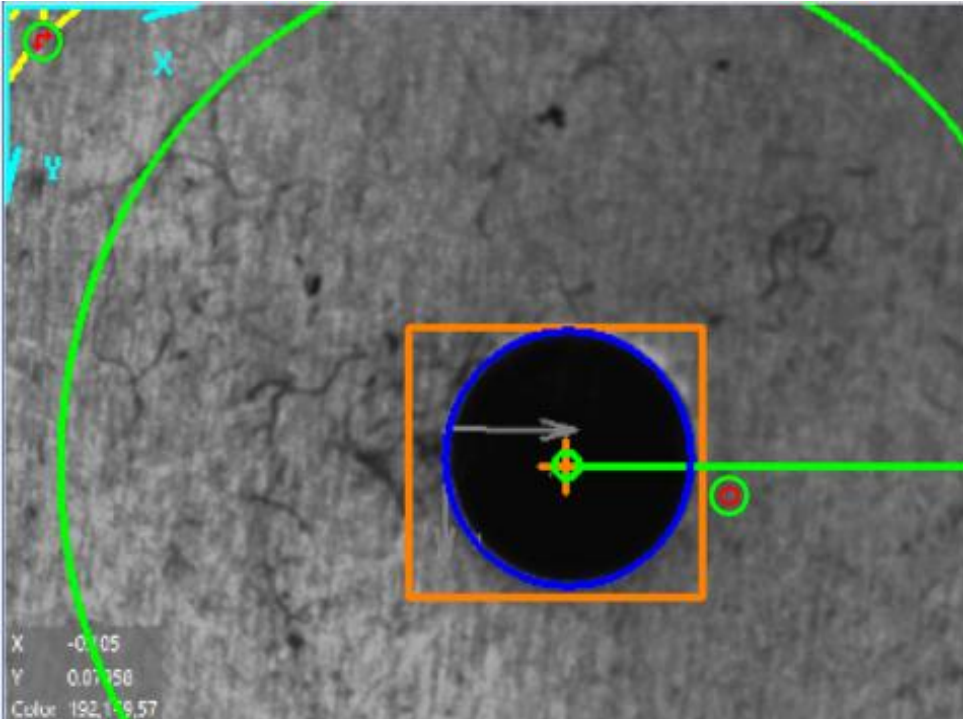


Figure 7: Example Measurement Process

Methodology & Results:

- Calibration of each camera
- Shape finding using color contrast
- Center-point recognition using found shape

Results							
Name	Status	Value	NumberFound	X	Y	Angle	Radius
GlobalFrame	Pass	1	0	0	0	0	
MeasuredCircle	Pass	1	445.338	346.845		97.1637	

Figure 8: Measurement Results

Test Type	Objective
Power System	Verify voltage stability and current availability during transient load spikes and sustained operation. Ensure components receive necessary power
Structural/Vibrational	Conduct mock inspections of system in baseline room AND active machine shop, confirming stability
Thermal	Evaluate the thermal stability of the electrical enclosure and ensure internal components operate within specified maximum temperatures
Data Integrity	Validate the Ignition interface logic and database communication reliability between devices

Successes	Remaining Issues
• Table and electrical box assembly • HMI and Database integration • Aurora flowchart logic	• Insufficient Ethernet switch • Oversized electrical box

Recommendations

Recommendations	
Electrical Box	Choose slower fan, downsize electrical box
HMI	Get more operator feedback, dynamically scale screen based on device resolution
Table	Have operators test the system and provide ergonomic feedback